

Contrastive Pretraining on Real LHC Jets: Transfer, Scorer Dependence and Mass Distortion

Animesh Parashar¹

Aditya Parashar²

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¹Department of Physics, IIT (ISM) Dhanbad, Dhanbad 826004, Jharkhand, India

²Department of Computer Science and Engineering, BIT Mesra, Ranchi 835215, India

Abstract

We compare contrastive pretraining on real AspenOpenJets and simulated LHC Olympics jets for background-based anomaly ranking, using matched random backbones and jet observables as controls. Equal training budgets comprise 100,000 jets per domain, one real-data shard and three seeds. Only 24 real-data training jets lie in the simulated reference sample’s central 90% transverse-momentum range, preventing a causal attribution to the pretraining domain. On partitions withheld from development, transverse-momentum balancing gives mean nearest-neighbor AUCs of 0.541, 0.564 and 0.517 for real-data, simulation and random representations on two-prong signals; three-prong values are 0.419, 0.553 and 0.426. All real-data seeds rank three-prong signal below background. Random backbones lead the original two-prong comparison with mean AUC 0.636. Signal–background momentum and multiplicity differences are consistent with score correlations and ranking changes, without establishing their cause. Regularized Mahalanobis and whitened nearest-neighbor scores do not make real-data representations competitive with girth; whitening benefits random features more. At background-calibrated operating points, girth retains more signal but strongly distorts background jet mass, whereas real-data-trained scores retain little signal. This bounded comparison characterizes representation, scorer and population dependence. Prior access to these public benchmarks limits independence; scorer and physics extensions are post-evaluation diagnostics.

1 Introduction and scope

Learning from real collision data is attractive when simulated backgrounds do not fully reproduce detector response and jet structure. However, a useful representation need not induce a useful anomaly ranking under a particular metric. A comparison of pretraining domains must therefore distinguish the representation, its readout, the evaluation population and optimization variability.

JetCLR established symmetry-informed contrastive jet representations and linear classifier tests [1]. Contrastive dijet representations and weakly supervised anomaly searches were investigated in Ref. [2]; AnomalyCLR uses anomaly-oriented augmentations and latent reconstruction scoring [3]. AspenOpenJets demonstrates beneficial real-data pretraining for generative transfer with OmniJet- α [4, 7]; that task differs from background-referenced anomaly ranking. OmniLearned demonstrates anomaly detection with CMS experimental data using large-scale multitask pretraining [5]. Metzger et al. show that representations optimal for background classification need not maximize new-physics sensitivity [6]; our readout-dependent findings are a bounded example of this distinction, not its first observation. We claim neither first discriminative use of AspenOpenJets nor a new contrastive objective. Related self-supervised approaches use re-simulation to define alternative views (RS3L) [19] or recover masked particles [20]. These extend the range of pretraining strategies; the present rotation-only comparison does not test them.

We ask a narrower question: under a specified equal-sample, equal-update recipe, does real-data pretraining improve simulated-signal ranking relative to simulation pretraining and matched random initialization? We then test how the answer depends on two established covariance-based scorers and on physically relevant operating points. Mass distortion matters because selection can reshape background spectra, a concern already studied in jet tagging and anomaly detection [8, 9]. Distance-correlation regularization provides an established route to reducing such dependence [21]; no mass-decorrelated control is implemented here. Our contribution is the combined empirical characterization of domain, seed, scorer and retention–distortion behavior in this particular transfer setting. The latter two analyses were specified after inspecting the fixed-score results. They broaden the characterization, not the independence of its evidence. The principal findings are:

1. The compared training distributions have little overlap in their dominant momentum ranges; the observed transfer result does not identify a domain cause.
2. No stable Aspen advantage emerges across three seeds; its three-prong raw-score ranking is inverted for every seed.
3. Whitening benefits matched random features more than Aspen features, while supervised probes reveal information not recovered by these anomaly scores.
4. Girth retains substantially more signal but strongly sculpts background jet mass; weak sculpting without signal retention is insufficient.

2 Datasets, representation and separation

2.1 Inputs and physical definitions

AspenOpenJets is derived from CMS 2016 Open Data [4]. We use the available `RunG_batch0.h5` shard and its stored jet constituents, with at most 150 constituents supplied per jet. The selected training sample contains 100,000 jets; it is unlabeled and not certified background-only. We do not use impact parameters, charge or constituent pileup weights in this comparison. The producer selects anti- k_t $R = 0.8$ jets with corrected $p_T > 300$ GeV and $|\eta| < 2.5$ from the JetHT stream, an inclusive combination of hadronic triggers, with standard event/jet quality requirements. Its jet reconstruction uses PUPPI and jet-energy corrections [4]. Our sums of stored, unweighted constituents are not those corrected producer jet quantities.

The LHC Olympics R&D sample [10, 11] contains one million QCD events and 100,000 two-prong signal events. Its $W' \rightarrow XY$ benchmark has masses $(m_{W'}, m_X, m_Y) = (3500, 500, 100)$ GeV and $X, Y \rightarrow qq$ decays. A separate 100,000-event sample uses $X, Y \rightarrow qqq$ with otherwise corresponding generator settings. The samples use Pythia8 and Delphes 3.4.1 without pileup or multiple-parton interactions and were selected with an $R = 1$ fat-jet trigger at $p_T > 1.2$ TeV [10]. Our subsequent $R = 0.8$ leading-jet analysis is a specified adaptation, not a reproduction of the producer’s $R = 1$ features.

For simulated events, massless particle inputs (p_T, η, ϕ) are converted to four-vectors and clustered with FastJet anti- k_t , $R = 0.8$, using E-scheme recombination [12, 13]. We choose the jet with largest vector p_T . Aspen inputs retain the producer’s stored constituent membership. For either domain, let $P^\mu = \sum_{i \in J} p_i^\mu$ over all available jet constituents and $S_T = \sum_{i \in J} p_{T,i}$. The axis is defined by

$$p_{T,J} = \sqrt{P_x^2 + P_y^2}, \quad \eta_J = \text{arsinh}(P_z/p_{T,J}), \quad \phi_J = \text{atan2}(P_y, P_x). \quad (1)$$

Before truncation, each constituent is represented as

$$x_i = \left(\eta_i - \eta_J, \text{wrap}_{[-\pi, \pi)}(\phi_i - \phi_J), \log \frac{p_{T,i}}{S_T}, \log \frac{p_{T,i}}{\text{GeV}} \right). \quad (2)$$

We retain the 50 highest- p_T constituents and explicitly mask padding. The full axis and denominator are not recomputed after truncation. Full-jet mass is $m_J = \sqrt{\max(0, P_\mu P^\mu)}$; full multiplicity counts available nonzero- p_T constituents before truncation. Shared feature definitions do not equalize selection, physics composition, pileup or detector response across domains. Only 3/100,000 selected Aspen training jets have 150 nonzero stored constituents. This measures saturation of the stored list, not a truth-level truncation-loss rate. Signal labels identify events rather than a truth-matched leading X or Y jet; Appendix A shows the leading-jet mass distributions.

2.2 Partitions and data exposure

Event-identity hashing assigns background or unlabeled events to training, validation, reference and test partitions with nominal fractions 60/15/10/15. Signals have only validation/test partitions, nominally 50/50. Events examined during initial data checks are assigned to validation. Aspen jets sharing an event identity share a partition. Training loaders reject signal, reference and test inputs. Both domains use fixed samples of 100,000 training and 8,192 validation jets; samples are identical across training seeds within each domain. Those checks covered 5120 Aspen jet rows from 3164 distinct events and 5120 LHCO events (4608 background and 512 signal). Their identities are reserved for validation, including other Aspen jets from an examined event.

Development evaluation uses 10,000 background validation references, 10,000 distinct background validation queries and 2,000 signal queries. The frozen final-partition evaluation uses 10,000 background reference events, 10,000 background test events and 2,000 signal test events per topology. Both final topologies share the same background events. Three-prong source IDs occupy a distinct namespace; 49,945/50,055 events enter validation/test, with no rejected rows and no overlap. Each simulated query is one leading jet per event.

Here “final” means withheld from the development of this pipeline. The public benchmarks were accessible before the final evaluation, so the final partitions do not provide fully independent confirmation. Development was adaptive, including inspection of signal-validation scores. The final protocol was fixed before the reported final-partition scoring, but subsequent scorer and physics extensions were motivated by those results. No independent discovery dataset is claimed.

2.3 Kinematic support of the training samples

After the primary evaluation, we reconstructed the exact training-row selections from the completed run manifests and compared them with the fixed background references (Fig. 1). The available-constituent vector- p_T medians are 336.8 GeV for Aspen training, 1310.9 GeV for simulation training and 1307.9 GeV for simulation references. Of 100,000 training jets, only 24 Aspen jets and 89,592 simulation jets lie in the reference central 90% range, [1198.6, 1740.9] GeV. All reference jets lie outside Aspen training’s central 98% range [257.5, 706.9] GeV. Central ranges describe concentration, not strict support; rare Aspen jets do extend into the evaluation regime.

The features explicitly retain the scalar momentum scale because $x_{i,4} - x_{i,3} = \log(S_T/\text{GeV})$. Evaluation-time balancing of signal and background does not match the pretraining domains. These measurements establish a major distribution mismatch, but do not determine how much of the ranking deficit it causes. Removing the absolute-momentum input would test explicit scale dependence without equalizing detector response, composition or all kinematic correlations.

3 Models, learning and scoring

3.1 Encoder and optimization

The backbone consists of a four-feature linear input map, four independently initialized pre-normalized Transformer encoder layers [14] with width 128, eight heads, feed-forward width 512, ReLU activations and dropout 0.1, followed by layer normalization and masked mean pooling.

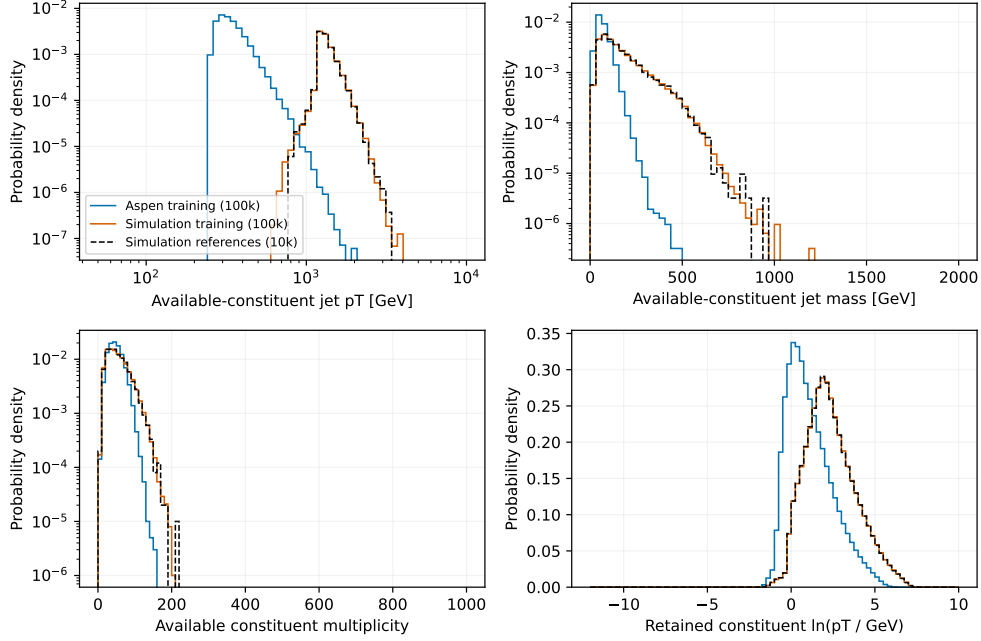


Figure 1: Exact selected training samples and fixed simulation references. Jet quantities use all available constituents; the constituent-momentum panel uses the retained 50 or fewer, with equal total weight per jet. Histograms retain their full-population normalization; underflow/overflow probabilities and particle-weighted alternatives are supplied in the numerical supplement.

Input features are divided by $(1, 1, 5, 5)$. With no positional encoding and masked mean pooling, the deterministic evaluation backbone is permutation-invariant up to numerical roundoff. A $128 \rightarrow 128 \rightarrow 256$ projection head with an intermediate GELU and final unit normalization is used only for training.

Two independently sampled local $(\Delta\eta, \Delta\phi)$ rotations, with angles uniform in $[-\pi, \pi]$, define positive views. The momentum features are unchanged. The implemented ϕ -chart rejection rule never triggers in the selected training samples: maximum local radii are 1.192 (Aspen) and 1.153 (simulation), both below π . This planar transformation approximates a narrow-jet augmentation; it is not an exact Lorentz/global-azimuth symmetry and need not preserve exact jet mass. No exact infrared/collinear safety is asserted for the finite, truncated network. Rotation-only views define a restricted recipe inspired by JetCLR, not a reproduction of its full augmentation study.

For $B = 128$ jets and $2B$ normalized projections u_i , paired-view index $p(i)$, the symmetric normalized temperature-scaled cross-entropy [15] is

$$\mathcal{L} = -\frac{1}{2B} \sum_{i=1}^{2B} \log \frac{\exp(u_i^T u_{p(i)}/\tau)}{\sum_{j \neq i} \exp(u_i^T u_j/\tau)}, \quad \tau = 0.1. \quad (3)$$

The positive is included in the denominator; only self-similarity is excluded. AdamW [16] uses learning rate 3×10^{-4} , weight decay 10^{-4} and gradient-norm clipping at 1. We train for 100 epochs without a learning-rate schedule and with dropped incomplete batches (781 updates/epoch). Validation views are fixed for loss monitoring; no signal-AUC endpoint selection is used. Seeds 17, 29 and 43 have matched initial backbones across domains. Random controls are their saved initial backbones, not separately tuned models. Seed 17 was continued from saved optimizer/RNG state to the fixed endpoint; the completed seed-43 runs use the original trainer from initialization after a walltime-limited attempt was discarded. Partial results are not selected or combined. The final fixed-view validation losses span 0.0208–0.0214 across the six runs; their histories appear in Appendix A. Similar endpoint losses do not establish similar representation or anomaly-ranking quality.

3.2 Frozen score and controls

Scoring uses raw 128-dimensional backbone outputs $h(x)$, not the normalized projection head. For background reference embeddings $\mathcal{R}$, the original score is

$$s_{10}(x) = \text{kth}_{10}\{\|h(x) - r\|_2 : r \in \mathcal{R}\}, \quad (4)$$

with larger scores designated anomalous. It is the tenth distance, not the mean of ten distances. AUCs below 0.5 are retained without reversing the score. Reference size 10,000 and $k = 10$ are fixed choices; their sensitivity is not tested.

Controls include high mass, high p_T , both signs of multiplicity, high girth, and tenth-neighbor distance in $(m_J, p_{T,J}, n_J, g_J)$ after background-reference mean/standard-deviation scaling. Girth is

$$g_J = \frac{\sum_{i \in J_{50}} p_{T,i} \sqrt{(\Delta\eta_i)^2 + (\Delta\phi_i)^2}}{\sum_{i \in J_{50}} p_{T,i}}, \quad (5)$$

using retained constituents relative to the full axis. It is not a full-jet girth. The observable-density control has access to full mass and multiplicity, so it is not an information-matched learned-representation comparison. Girth is computable from the encoder inputs. These controls test practical value rather than claim a matched optimization of all possible physics taggers.

3.3 Subsequent covariance-scorer extension

For each backbone, background references alone fit

$$\mu = \frac{1}{N} \sum r_i, \quad C = \frac{1}{N} \sum (r_i - \mu)(r_i - \mu)^T, \quad C_\lambda = 0.9C + 0.1 \frac{\text{tr } C}{128} I. \quad (6)$$

With $C_\lambda = LL^T$, column vectors transform as $z = L^{-1}(h - \mu)$. We evaluate $s_M = \|z\|^2$ and tenth-neighbor distance in transformed coordinates. Regularization, k , models and score signs are fixed before this extension; there is no hyperparameter search. The regularized covariance, not necessarily the empirical covariance, whitens to identity. Mahalanobis distance is not a calibrated tail probability. All 27 backbone/scorer combinations per topology are reported. The extension is descriptive post-evaluation robustness analysis.

4 Evaluation and uncertainty

4.1 AUC and kinematic support

We report original-population AUC alongside a common- p_T diagnostic. Twenty background-reference quantile bins define edges. Bins must contain at least 30 queries of each class and lie within reference support. For class c in supported bin b , the event weight is

$$w_i = \frac{\min(n_{0b}, n_{1b})}{n_{cb}}, \quad i \in (c, b); \quad w_i = 0 \text{ otherwise.} \quad (7)$$

Weighted AUC is the weighted signal-background pair average of $\mathbf{1}[s_S > s_B] + \frac{1}{2}\mathbf{1}[s_S = s_B]$, normalized by the product of class weight sums. This equalizes coarse p_T -bin class masses, not within-bin kinematics or other confounders. It uses evaluation labels and is not a deployment selection. The final comparison excludes 5,468/10,000 background queries and 186/2,000 two-prong or 180/2,000 three-prong signals. Effective sizes $(\sum w)^2 / \sum w^2$ are approximately 2,329/1,789 and 2,371/1,775 for the two background/signal pairs. Thus balanced and original AUCs describe materially different populations and must be presented together. For both topologies, supported bins are reference quantile bins 12–20 (one-based), spanning approximately 1327.1–3223.1 GeV; exact edges and counts are supplied with the diagnostic results.

Three-seed arithmetic means and sample SDs refer to individual-model metrics, not averaged-score ensembles. Domain differences are paired by seed. Shared, class-stratified event bootstraps (1,000 replicates) recompute the balancing weights with fixed edges. Their percentile intervals condition on fitted models and reference events; they exclude training and reference-fitting uncertainty. They cannot substitute for seed variability. Later extensions use pointwise intervals without multiplicity-adjusted confirmatory claims. Failure to establish an advantage is not evidence of equivalence.

4.2 Background-calibrated physics diagnostic

The 10,000 shared background queries are split once into disjoint 5,000-event calibration and assessment sets. For target background acceptance $a \in \{0.10, 0.05\}$, the threshold is the $\lceil 5000a \rceil$ -th largest calibration score; selections include all ties. All 2,000 signals of each topology assess efficiency. No signal enters threshold fitting. This uses original, unweighted populations, and actual assessment acceptance is reported rather than assumed equal to the target.

Let F and F_{sel} be inclusive and selected assessment-background mass CDFs. We measure $D_{\text{KS}} = \sup_m |F(m) - F_{\text{sel}}(m)|$ without an independent-sample KS p -value: selected events are a subset of the inclusive sample. We also use

$$\text{JSD}(p, q) = \frac{1}{2} \sum_b p_b \log \frac{p_b}{(p_b + q_b)/2} + \frac{1}{2} \sum_b q_b \log \frac{q_b}{(p_b + q_b)/2}, \quad (8)$$

for normalized ten-bin mass histograms. Bins are calibration-background quantiles, with outer edges extended to infinity, no pseudocounts, and $0 \log 0 = 0$. JSD is in natural-log units. Empty selected samples would yield undefined shape metrics, not zero distortion. Marginal acceptance profiles use analogous mass and p_{T} bins. They do not measure mass dependence conditional on p_{T} .

Finite samples create apparent distortion even for mass-neutral selection. Five hundred uniform subsets of the assessment background, matched to each observed selected count, provide descriptive KS/JSD reference bands. These are not discovery p -values. Conditional binomial uncertainty uses Wilson intervals [17]; 1,000 paired bootstraps additionally resample calibration background, assessment background and each signal class independently and refit thresholds in every replicate. The selected/inclusive nesting and ties are preserved. Models, reference fits, bins and the event split remain fixed. Both topologies share background events and are not independent experimental confirmations.

5 Results

5.1 Transfer and seed variation

Table 1 and Fig. 2 show the frozen final-partition comparison. Two-prong common- p_{T} Aspen-minus-simulation differences are $-0.0789, +0.0965, -0.0872$ for seeds 17, 29 and 43. Their mean is -0.0232 with sample SD 0.1038; the fixed-model event-bootstrap interval $[-0.0448, -0.0026]$ conditions on these three trained models and the reference events. It excludes training variability and therefore is not evidence for a reproducible domain difference. No stable domain advantage is established. In the original population, high p_{T} alone has AUC 0.833 (two-prong) and 0.835 (three-prong), exceeding every learned score in this comparison. After common- p_{T} balancing its AUC is approximately 0.494–0.495, as expected when coarse momentum differences are removed. For three-prong signals all three Aspen raw-score balanced AUCs are below 0.5. This is inverted ranking under the prescribed score direction, not proof that all signal information is absent. Girth exceeds every trained seed in the balanced population on both topologies.

Score	2p original	2p balanced	3p original	3p balanced
Aspen	0.5114 ± 0.0626	0.5407 ± 0.0391	0.3968 ± 0.0534	0.4188 ± 0.0317
Simulation	0.5504 ± 0.2170	0.5639 ± 0.0672	0.5446 ± 0.2219	0.5528 ± 0.0848
Random	0.6361 ± 0.0164	0.5172 ± 0.0174	0.5580 ± 0.0168	0.4255 ± 0.0160
High mass	0.6390	0.6065	0.6637	0.6322
High p_T	0.8334	0.4940	0.8353	0.4948
High multiplicity	0.4265	0.4301	0.5744	0.5770
Low multiplicity	0.5735	0.5699	0.4256	0.4230
Girth	0.6858	0.7131	0.6838	0.7118
Observable density	0.8095	0.6452	0.7691	0.6060

Table 1: Primary frozen-score final-partition AUCs; 2p and 3p denote two-prong and three-prong signals. Model entries are mean $\pm$ sample SD over three training/initialization seeds, not confidence intervals. Physics controls have no encoder seeds. “Balanced” denotes the restricted common- p_T population.

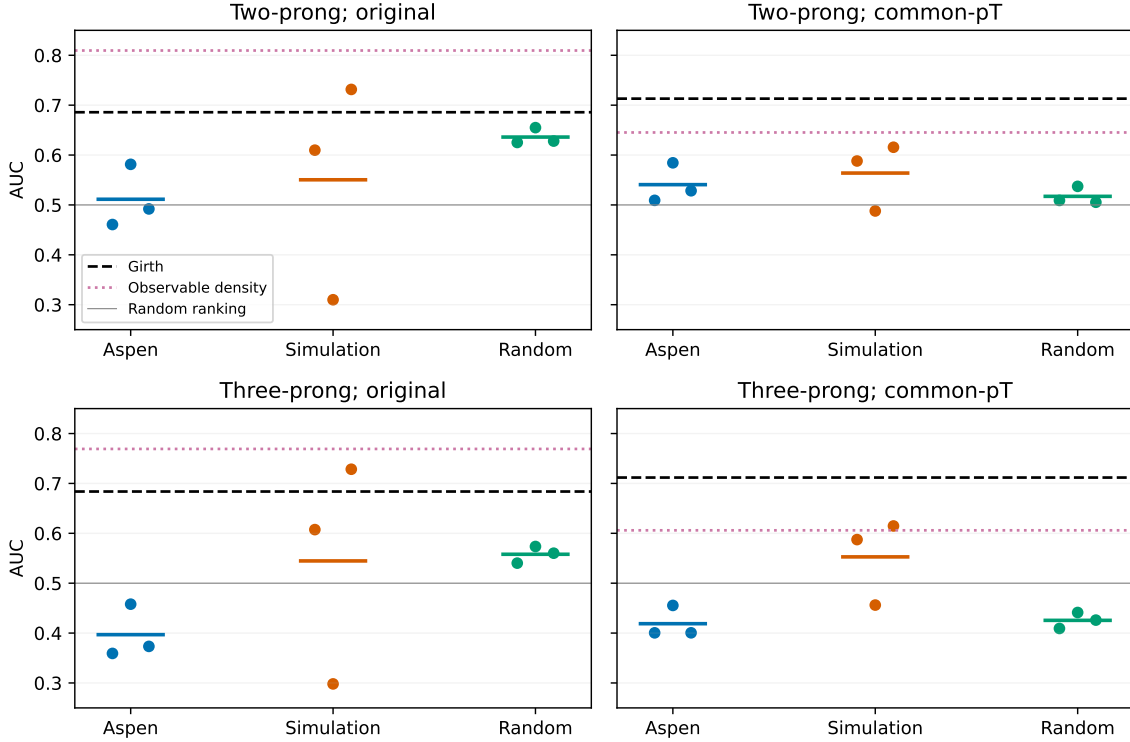


Figure 2: Each point is one seed; short bars show seed means. Dashed/dotted lines are physics controls. Both the population definition and initialization materially change the apparent ranking. No seed is discarded.

5.2 Dependence on the scorer

The subsequent scorer comparison is given in Table 2 and Fig. 3. Whitening changes Aspen common- p_T mean AUC by $+0.0019$ (two-prong) and -0.0087 (three-prong), compared with $+0.0470$ and $+0.0478$ for random backbones. On two-prong queries, the Aspen-minus-random difference changes from $+0.0235$ under raw kNN to -0.0217 under whitened kNN and -0.0281 under Mahalanobis scoring. An improvement after metric adjustment cannot therefore be attributed to learning without the corresponding random-feature control. Simulation improves modestly under whitening, with substantial seed variability. Every trained seed/scorer remains below girth in both balanced populations. These findings extend the weak result beyond raw Euclidean kNN but do not test all anomaly scores or configurations.

Backbone	Scorer	2p original	2p balanced	3p original	3p balanced
Aspen	Raw kNN	0.5114 ± 0.0626	0.5407 ± 0.0391	0.3968 ± 0.0534	0.4188 ± 0.0317
Aspen	Mahalanobis	0.4664 ± 0.0353	0.5182 ± 0.0097	0.3678 ± 0.0330	0.4162 ± 0.0137
Aspen	Whitened kNN	0.5078 ± 0.0532	0.5426 ± 0.0225	0.3843 ± 0.0527	0.4101 ± 0.0233
Simulation	Raw kNN	0.5504 ± 0.2170	0.5639 ± 0.0672	0.5446 ± 0.2219	0.5528 ± 0.0848
Simulation	Mahalanobis	0.5555 ± 0.1886	0.5591 ± 0.0706	0.5518 ± 0.2017	0.5596 ± 0.0908
Simulation	Whitened kNN	0.5653 ± 0.1962	0.5849 ± 0.0711	0.5538 ± 0.2054	0.5698 ± 0.0904
Random	Raw kNN	0.6361 ± 0.0164	0.5172 ± 0.0174	0.5580 ± 0.0168	0.4255 ± 0.0160
Random	Mahalanobis	0.6366 ± 0.0086	0.5463 ± 0.0106	0.5598 ± 0.0070	0.4623 ± 0.0123
Random	Whitened kNN	0.6296 ± 0.0141	0.5642 ± 0.0132	0.5420 ± 0.0094	0.4733 ± 0.0100

Table 2: All fixed scorers on original and common- p_T populations, mean $\pm$ seed SD. This comparison was added after the primary final-score evaluation.

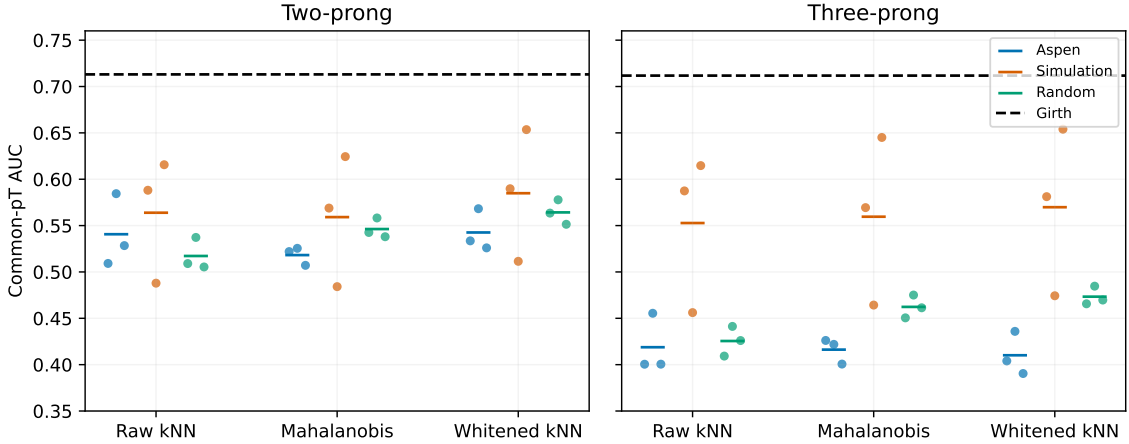


Figure 3: Common- p_T scorer sensitivity. Short horizontal markers show seed means; offset points show every seed. Scorers are discrete fixed choices.

5.3 Signal retention versus mass distortion

At nominal 5% background acceptance, raw Aspen has actual mean acceptance 5.53%, retains 3.17%/0.92% of two-/three-prong signals, and has mean mass KS distance 0.135. Girth has actual acceptance 4.82%, retains 24.10%/22.05%, and has KS 0.893. Observable density has acceptance 4.74%, retains 31.85%/30.30%, and has KS 0.586. Raw simulation lies between these examples with 4.95% acceptance, 9.75%/9.68% retention and KS 0.318, but its signal-efficiency seed SDs are 9.15/8.59 percentage points. Actual acceptance differences and training variation must accompany the comparison.

Girth’s stronger retention comes with pronounced mass sculpting. Its 241 selected assessment-background events at nominal 5% give KS 0.893, versus a count-matched random-selection reference range approximately [0.029, 0.093]. Conversely, Aspen’s weaker distortion comes with mean signal acceptance below mean background acceptance for every tested scorer and both operating points. A signal-independent random selection would retain equal fractions in expectation. Weak sculpting alone does not establish a useful anomaly tagger. Neither spectrum distortion nor these efficiencies establish a fake resonance, sideband-fit bias or discovery significance. The measured mass is jet mass, not dijet mass. KS is a marginal diagnostic and includes distortion induced by p_T selection. No mass-decorrelated tagger or optimal retention–distortion frontier is claimed.

Raw-score backbone	Selected background counts (seeds 17, 29, 43)	Total
Aspen	269, 284, 277	5000
Simulation	238, 246, 258	5000
Random	267, 259, 264	5000

Table 3: Raw-score selected assessment-background counts at nominal 5%. Girth selects 241 and observable density 237 of the same 5000 events. All scorer counts and achieved rates are supplied in the numerical supplement.

5.4 Representation geometry and score correlations

We use the cached embeddings of all nine frozen backbones to characterize their geometry and their dependence on elementary jet observables. On 10,000 background queries, raw Aspen scores anticorrelate with available multiplicity (Spearman $\rho = -0.53$ to -0.58). Simulation’s score– p_T correlation changes from -0.436 for seed 29 to $+0.289$ for seed 43. These associations are consistent with sensitivity to jet-level properties, and the evaluated populations clarify their possible relevance (Fig. 6). Original signal momentum medians are 1645.7 and 1644.7 GeV for two- and three-prong jets, compared with 1309.7 GeV for background. Increasing momentum alone gives AUCs 0.833 and 0.835, falling to 0.494 and 0.495 after the existing coarse-bin balancing. The harder signals are consistent with the ordering of simulation seeds 29 and 43 by their score–momentum correlation and original two-prong AUC (0.31 versus 0.73). This is an association, not a decomposition of the learned score.

Multiplicity differs in opposite directions: original medians are 41 for two-prong signal, 49 for background and 53 for three-prong signal. After momentum balancing, increasing multiplicity gives AUCs 0.430 and 0.577, respectively. Because larger Aspen scores denote more anomalous jets, its negative background score–multiplicity correlation predicts a multiplicity contribution toward two-prong AUC above 0.5 and three-prong AUC below 0.5. The observed balanced means, 0.541 and 0.419, have those signs; this correlation alone does not determine their magnitudes. Background-only correlations and marginal distributions cannot establish that either observable causes the learned-score inversion.

For background-reference covariance eigenvalues λ_j , define $p_j = \lambda_j / \sum_k \lambda_k$ and entropy effective rank $r_{\text{eff}} = \exp[-\sum_{p_j > 0} p_j \log p_j]$. Aspen ranks are 6.11–7.65, simulation 15.25–17.10 and random 3.20–3.99 (Fig. 7). Both trained families have higher effective rank than their random controls. Concentrated variance alone neither establishes pathological collapse nor explains the differential whitening gains. Dimensional collapse can occur in contrastive learning [22], but an effective rank below 128 is not, by itself, evidence of that training pathology. Reference/query spectra, participation ranks and correlations for all scorers are supplied, without correlation-significance claims.

A complementary single-seed development check gives mean five-fold supervised linear-probe balanced AUCs 0.8516 (Aspen), 0.8691 (simulation) and 0.8036 (random). Thus information

Signal retention versus background jet-mass distortion

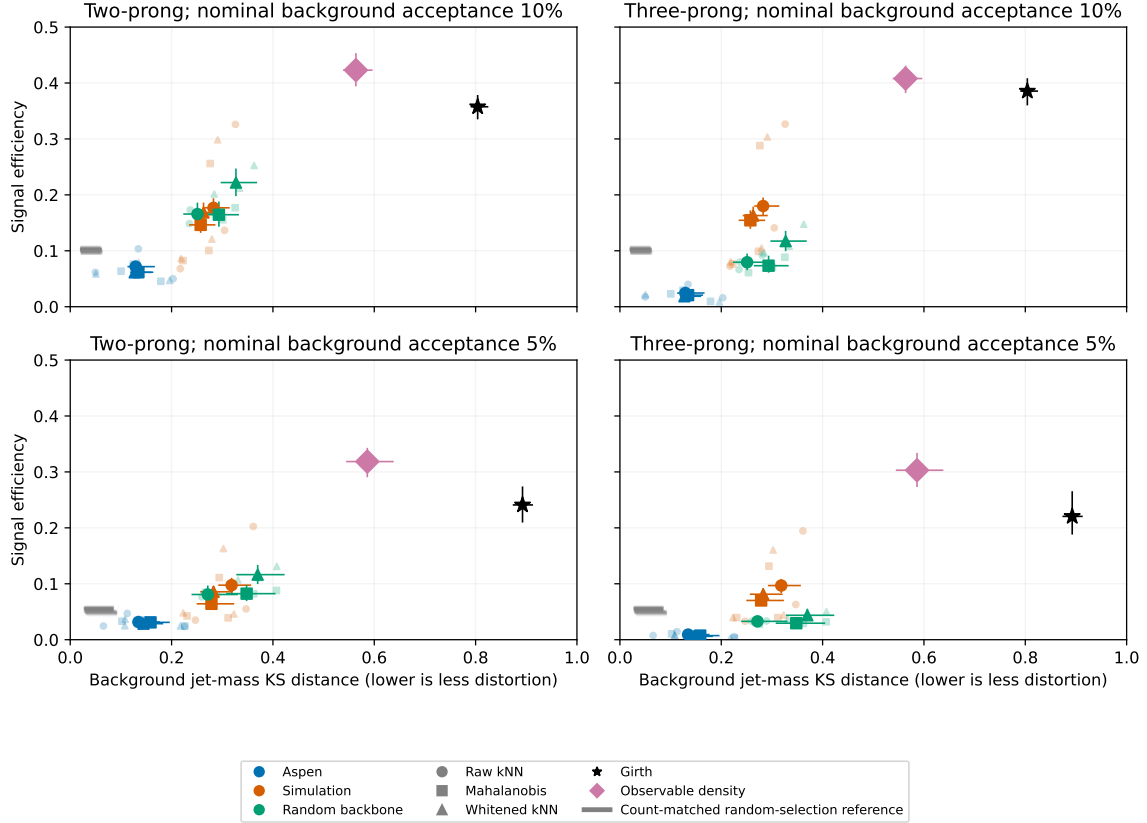


Figure 4: Signal retention versus background jet-mass distortion. Large model markers show three-seed means and faint points show individual seeds; shape identifies the scorer. Horizontal/vertical segments are marginal 95% event-bootstrap intervals with threshold refitting, conditional on trained models and references; they are not joint regions or training uncertainty. Gray segments show each arm’s count-matched random-selection KS band at its achieved background acceptance, the expected signal efficiency of an independent random selection. These are reference expectations, not measured random-cut signal efficiencies. All values use unweighted populations; nominal rates are in titles and achieved rates in Appendix D.

accessible to a label-trained readout can coexist with weak background-only kNN ranking. These probes use labels and do not establish unsupervised discovery performance; details are in the appendix. The three-prong inversion means that signal tends to receive smaller distances to the background reference set under this metric. It is not a statement about physical probability density, and reversing the score after seeing labels is not an independent remedy.

6 Limitations

This study compares two concrete data distributions. Domain identity is confounded with physics mixture, selection, detector response, pileup and kinematic support; a causal effect of “realness” is not identified. Section 2.3 quantifies the large momentum mismatch but does not remove it. It uses one real-data shard, 100,000 training jets per domain, three seeds, two related simulated topologies, one architecture and a small fixed set of scorers. It is not a result about all real-data pretraining, foundation models or contrastive objectives.

Momentum-matched pretraining or a controlled scale-feature ablation would be needed to

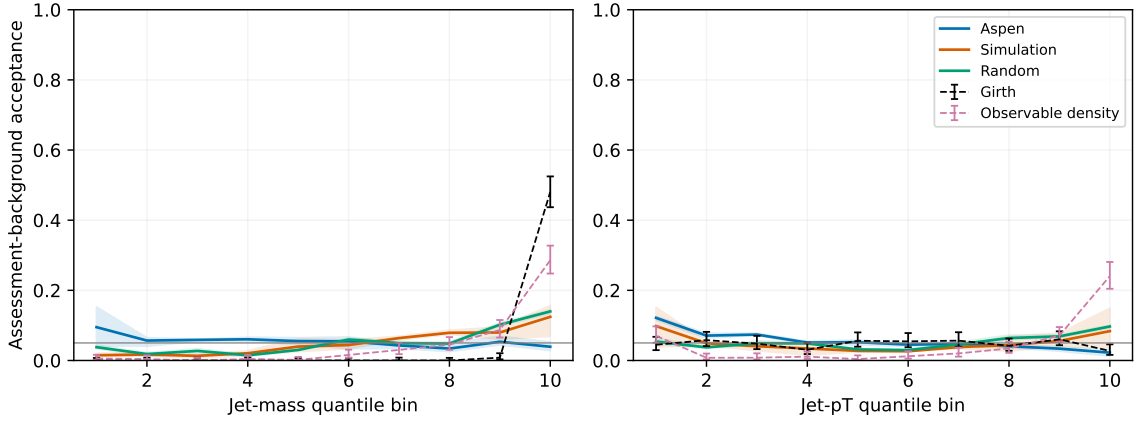


Figure 5: Background acceptance versus calibration-defined mass and p_T quantile bins at nominal 5%. Raw-score curves show seed means and shaded seed minima/maxima, not confidence bands. Controls show fixed-threshold 95% Wilson intervals. Outer bins include overflow. These marginal profiles do not isolate mass effects at fixed p_T . Conditional binomial intervals for all profiles are supplied in the numerical supplement.

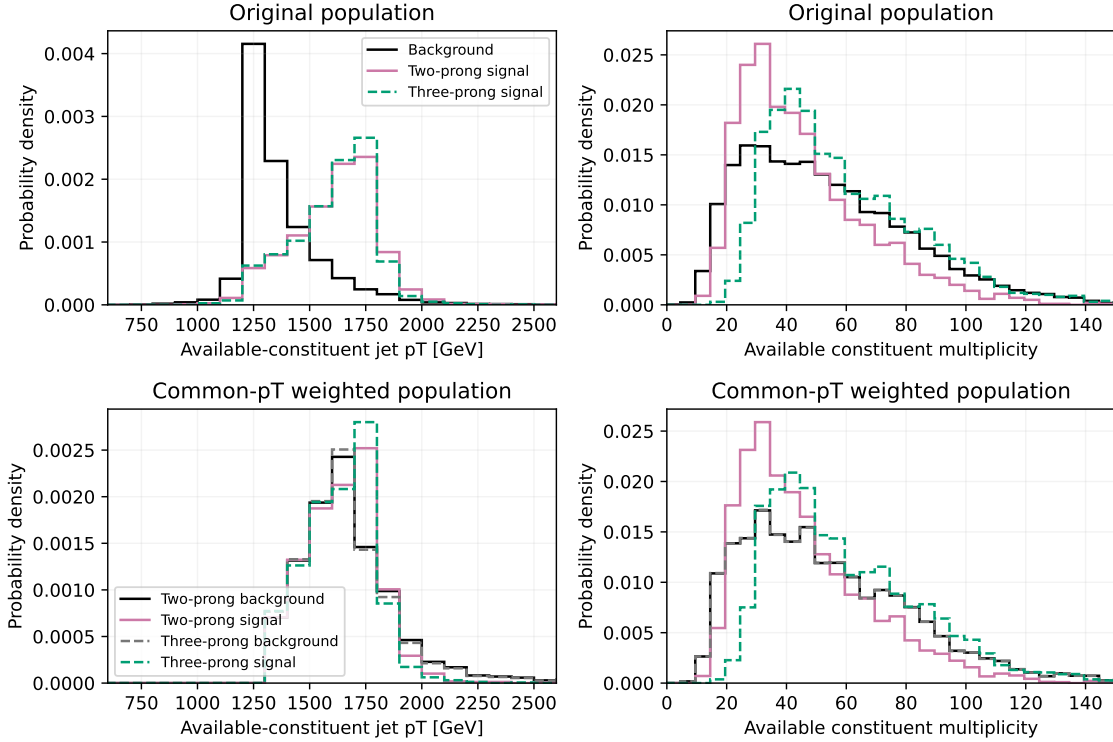


Figure 6: Momentum and available constituent multiplicity for the already-evaluated queries, before and after the existing common-momentum weighting. Original background events are shared across topologies; balanced background weights depend on the signal topology. Histograms use full-population normalization, including any probability outside the displayed range in the numerical supplement. No new selection or model tuning is performed.

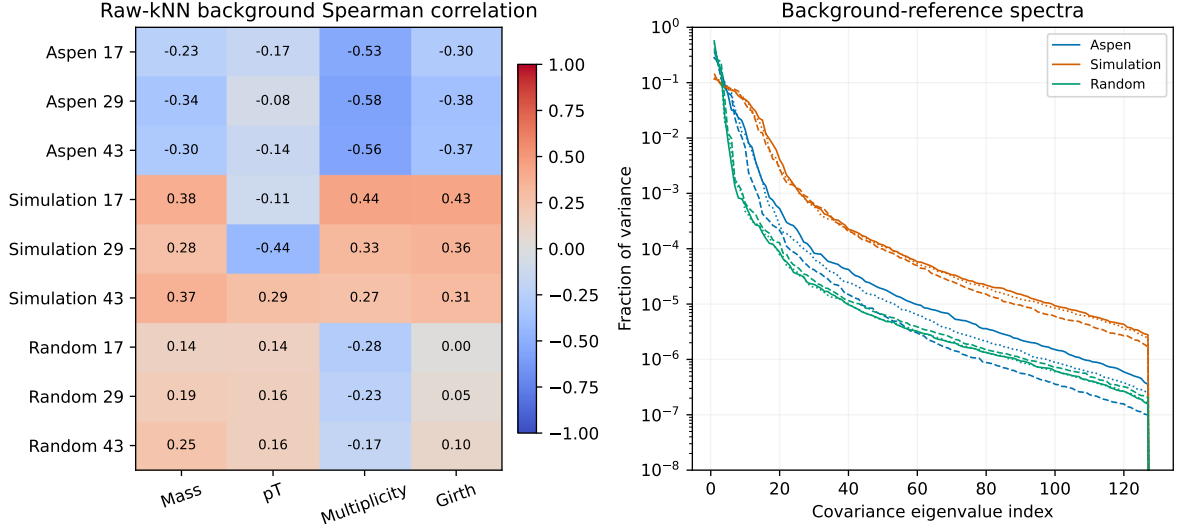


Figure 7: Post-hoc raw-score background correlations and reference covariance spectra for every seed. Spectral line styles indicate seeds 17 (solid), 29 (dashed) and 43 (dotted). Eigenvalues are normalized within each model; display range is limited to 10^{-8} –1, while full spectra are supplied.

test the role of the measured mismatch. Extrapolating the 24/100,000 overlap fraction to the full Aspen collection assumes shard representativeness, which has not been established. Counting the full available training shard yields only 350 jets from 243 events in the reference central 90% momentum range; a large, event-independent matched sample requires additional data. More seeds, a full JetCLR augmentation comparison and a mass-decorrelated girth control remain untested extensions.

Development and prior access to public benchmarks limit independence. The scorer, operating-point and representation studies were performed after the primary evaluation and are descriptive. Event intervals condition on fixed models/reference fits; three seeds give limited information about optimization variability. The paired two-prong domain-difference seed SD of 0.104 does not permit a precise conclusion about differences of a few hundredths with only three seeds; no equivalence or post-hoc power claim is made. No multiple-comparison discovery claim is made. The mass-distortion analysis does not implement a background-estimation procedure or experimental resonance search. A negative result in this setting is informative only within these limits.

7 Conclusion

Equal-budget contrastive pretraining on real jets does not demonstrate a robust anomaly-ranking advantage in this transfer configuration of 100,000 jets per domain, one real-data shard and three seeds. The training momentum mismatch precludes attributing this result to realness alone. Domain rankings vary with seed, and matched random controls change the interpretation of covariance scoring gains. Physics observables attain stronger signal retention but can strongly reshape background jet mass; reduced distortion of the Aspen scores comes with poor retention. The contribution is a reproducible, bounded comparison of representation, scorer, population and operating-point behavior. These results motivate careful evaluation of transfer representations without implying a new anomaly-detection method or a validated search.

Data and computational artifacts

The source datasets are public [4, 10]. The accompanying reproducibility package contains the numerical tables, machine-readable result summaries, frozen protocols, configuration files, analysis source and provenance manifests. The full raw data and model checkpoints are not embedded in the manuscript source archive. The code and configuration archive is prepared for release; a public repository for this version and model checkpoints are not yet available. Reproduction instructions identify the required source files and computational stages; exact trained-state reproduction can depend on the GPU/software environment. Saved scores reproduce the numerical tables and figures; regenerating model scores requires the recorded checkpoints or retraining with the documented procedure.

A Supporting training and representation checks

Table 4 reports all primary background correlations and effective ranks. The leading-jet signal mass plot describes reconstructed populations, not a truth assignment to X or Y . Fixed-view validation histories show the recorded endpoints, not an AUC-based stopping criterion. These checks were performed after the final evaluation and are interpreted descriptively.

Backbone	Seed	ρ_m	ρ_{p_T}	ρ_n	ρ_g	r_{eff}
Aspen	17	-0.234	-0.169	-0.532	-0.302	7.54
Aspen	29	-0.340	-0.075	-0.578	-0.381	6.11
Aspen	43	-0.305	-0.143	-0.557	-0.366	7.65
Simulation	17	0.384	-0.114	0.445	0.431	17.10
Simulation	29	0.280	-0.436	0.327	0.358	15.25
Simulation	43	0.370	0.289	0.270	0.309	16.01
Random	17	0.137	0.144	-0.278	0.003	3.20
Random	29	0.187	0.158	-0.231	0.054	3.99
Random	43	0.245	0.161	-0.172	0.101	3.33

Table 4: Raw-kNN background Spearman correlations and background-reference entropy effective rank. n is available multiplicity and g retained-constituent girth.

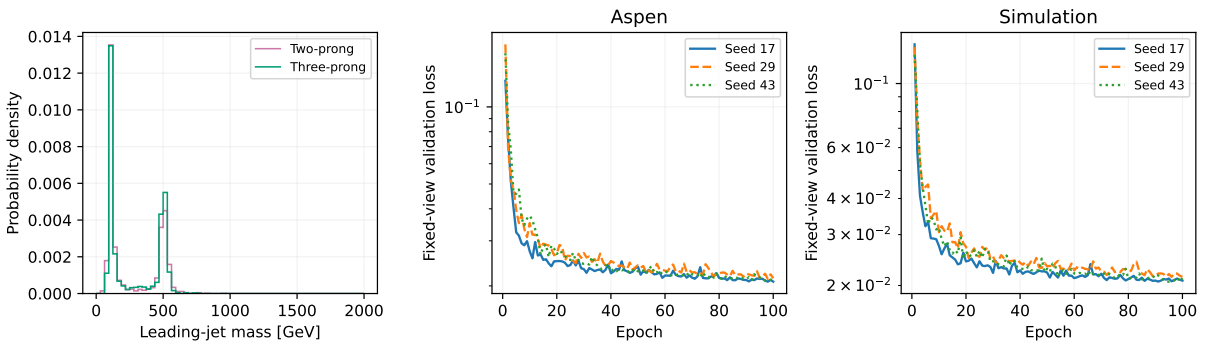


Figure 8: Already-scored leading signal jet masses and recorded fixed-view validation losses for the six trained models. No truth X/Y fraction is inferred.

B Additional development results

Development raw kNN mean common- p_T AUCs were 0.5217 ± 0.0388 for Aspen, 0.5562 ± 0.0737 for simulation and 0.5139 ± 0.0162 for random features. Single-seed supervised probes gave

mean five-fold balanced AUCs 0.8516, 0.8691, 0.8036 and 0.8502 for Aspen, simulation, random and four-observable inputs. Five shared stratified folds used seed 20261002. Per-fold feature standardization and background-reference-defined class-balancing weights were fitted on training folds only; scikit-learn logistic regression [18] used L2 regularization $C = 1$, an unpenalized intercept, LBFGS tolerance 10^{-8} and at most 2,000 iterations. Held-out-fold predictions defined fold AUCs. These label-using readouts diagnose accessible information, not background-only anomaly sensitivity. Fold spreads are not independent training uncertainty.

A 200-epoch single-seed Aspen constituent autoencoder gave development balanced AUCs about 0.494 for latent kNN and 0.474 for reconstruction error. It is contextual, not a budget-matched objective comparison. Positive collinear-splitting and soft angular-noise ablations, and a single-seed metric screen, did not justify further compute under development triage rules. Such rules do not statistically reject the usefulness of these methods. These exploratory arms are not selected for the primary final comparison.

C Deterministic sampling and numerical checks

Hash partitions use BLAKE2b with namespace, event identity and seed 20260925; the 64-bit digest modulo 10,000 determines the partition. Initially inspected events enter validation only. Source-row qq identities are negative one minus row index, in a separate namespace. Training sampling seed is 20260925. Final sample indices are sorted without-replacement choices from each partition with NumPy `default_rng(20261004)`. Physics calibration/assessment indices use `default_rng(20261006).permutation(10000)`. Its threshold-refitting bootstrap uses seed 20261007 and random-selection reference bands seed 20261008. Primary final AUC bootstraps use seed 20261003; subsequent scorer bootstraps use 20261005.

Source and checkpoint SHA256 manifests, completed-epoch histories, finite weights, paired initializations, fixed sample provenance and partition separation were checked. Raw scores were reproduced from cached embeddings. Independent float64 distances and linear-system quadratic forms checked covariance scoring; rank-based and weighted-pair calculations checked AUC. Physics checks reproduced thresholds, selected counts, KS and JSD with independent implementations, including ties and bootstrap duplicates. Saved arrays retain all 1,000 bootstrap replicates. Numerical checks establish implementation consistency, not protection against all scientific misspecification.

D Complete seed and operating-point tables

Backbone	Scorer	Seed	2p original	2p balanced	3p original	3p balanced
Aspen	Raw kNN	17	0.4607	0.5092	0.3592	0.4005
Aspen	Raw kNN	29	0.5814	0.5844	0.4579	0.4554
Aspen	Raw kNN	43	0.4920	0.5284	0.3733	0.4005
Aspen	Mahalanobis	17	0.4412	0.5220	0.3475	0.4261
Aspen	Mahalanobis	29	0.5068	0.5255	0.4059	0.4219
Aspen	Mahalanobis	43	0.4513	0.5071	0.3499	0.4006
Aspen	Whitened kNN	17	0.4704	0.5335	0.3484	0.4040
Aspen	Whitened kNN	29	0.5686	0.5682	0.4448	0.4359
Aspen	Whitened kNN	43	0.4842	0.5260	0.3596	0.3904
Simulation	Raw kNN	17	0.6098	0.5881	0.6074	0.5874
Simulation	Raw kNN	29	0.3099	0.4879	0.2981	0.4561
Simulation	Raw kNN	43	0.7315	0.6156	0.7284	0.6147
Simulation	Mahalanobis	17	0.5950	0.5689	0.5889	0.5694
Simulation	Mahalanobis	29	0.3502	0.4841	0.3342	0.4643
Simulation	Mahalanobis	43	0.7213	0.6244	0.7325	0.6451
Simulation	Whitened kNN	17	0.6021	0.5898	0.5889	0.5812
Simulation	Whitened kNN	29	0.3533	0.5115	0.3331	0.4743
Simulation	Whitened kNN	43	0.7404	0.6535	0.7393	0.6540
Random	Raw kNN	17	0.6252	0.5090	0.5402	0.4092
Random	Raw kNN	29	0.6549	0.5372	0.5736	0.4413
Random	Raw kNN	43	0.6280	0.5054	0.5601	0.4260
Random	Mahalanobis	17	0.6372	0.5426	0.5551	0.4505
Random	Mahalanobis	29	0.6450	0.5582	0.5678	0.4751
Random	Mahalanobis	43	0.6277	0.5380	0.5564	0.4613
Random	Whitened kNN	17	0.6315	0.5635	0.5382	0.4656
Random	Whitened kNN	29	0.6426	0.5778	0.5526	0.4846
Random	Whitened kNN	43	0.6147	0.5514	0.5351	0.4697

Table 5: All model/scorer seed AUCs. The corresponding CSV preserves numerical precision. The balanced columns use the restricted common- p_T population.

Backbone	Scorer	BG (%)	2p (%)	3p (%)	KS	JSD
Aspen	Raw kNN	10.247 ± 0.121	7.167 ± 2.816	2.450 ± 1.344	0.129 ± 0.077	0.015 ± 0.013
Aspen	Mahalanobis	10.093 ± 0.551	6.167 ± 1.533	2.067 ± 1.020	0.135 ± 0.040	0.018 ± 0.007
Aspen	Whitened kNN	9.973 ± 0.050	6.150 ± 1.621	1.867 ± 0.909	0.127 ± 0.073	0.017 ± 0.013
Simulation	Raw kNN	9.920 ± 0.352	17.683 ± 13.365	18.000 ± 13.141	0.283 ± 0.058	0.055 ± 0.020
Simulation	Mahalanobis	10.053 ± 0.281	14.633 ± 9.540	15.450 ± 11.616	0.257 ± 0.030	0.048 ± 0.015
Simulation	Whitened kNN	9.707 ± 0.167	16.850 ± 11.394	16.300 ± 12.229	0.263 ± 0.039	0.050 ± 0.015
Random	Raw kNN	10.167 ± 0.031	16.567 ± 1.492	7.950 ± 1.276	0.251 ± 0.026	0.045 ± 0.009
Random	Mahalanobis	10.393 ± 0.160	16.450 ± 1.117	7.333 ± 1.397	0.293 ± 0.037	0.069 ± 0.015
Random	Whitened kNN	10.353 ± 0.196	22.200 ± 2.693	11.733 ± 2.675	0.327 ± 0.040	0.076 ± 0.016
Girth	–	10.260	35.750	38.550	0.804	0.428
Observable density	–	10.300	42.300	40.800	0.564	0.216

Table 6: All methods at nominal 10% background acceptance. Efficiencies are percentages; model rows show mean $\pm$ seed SD. KS and JSD concern background jet mass; JSD uses natural logs. Actual acceptance is measured on a disjoint background assessment sample.

Backbone	Scorer	BG (%)	2p (%)	3p (%)	KS	JSD
Aspen	Raw kNN	5.533 ± 0.150	3.167 ± 1.329	0.917 ± 0.473	0.135 ± 0.082	0.018 ± 0.017
Aspen	Mahalanobis	5.560 ± 0.223	3.100 ± 0.577	0.733 ± 0.388	0.158 ± 0.063	0.023 ± 0.013
Aspen	Whitened kNN	5.507 ± 0.064	2.867 ± 0.679	0.600 ± 0.304	0.145 ± 0.063	0.019 ± 0.013
Simulation	Raw kNN	4.947 ± 0.201	9.750 ± 9.148	9.683 ± 8.590	0.318 ± 0.062	0.068 ± 0.022
Simulation	Mahalanobis	4.880 ± 0.347	6.417 ± 4.060	7.017 ± 5.312	0.279 ± 0.043	0.058 ± 0.020
Simulation	Whitened kNN	4.993 ± 0.081	8.567 ± 6.698	8.133 ± 6.860	0.283 ± 0.053	0.058 ± 0.025
Random	Raw kNN	5.267 ± 0.081	8.083 ± 0.451	3.283 ± 0.076	0.271 ± 0.016	0.055 ± 0.009
Random	Mahalanobis	5.507 ± 0.076	8.233 ± 0.551	2.950 ± 0.229	0.348 ± 0.067	0.090 ± 0.027
Random	Whitened kNN	5.693 ± 0.150	11.633 ± 1.286	4.367 ± 0.679	0.370 ± 0.038	0.090 ± 0.016
Girth	–	4.820	24.100	22.050	0.893	0.504
Observable density	–	4.740	31.850	30.300	0.586	0.244

Table 7: As above at nominal 5%. Full per-seed thresholds, counts, Wilson intervals, threshold-refitting bootstrap intervals, mass/ p_T profiles and paired differences are supplied in the machine-readable numerical supplement.

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